

(A06)CUSTOMER:

(B01) STAINLESS STEEL FORGED FLANGES

SO 9901, 2015 TÜV NORD REG NO 0410001111 TO EXPIRY DATE: 21.01.2019 & APPROVED ACCORDING TO AD 2000 MERKBLATT Wm & CERTIFIED ACCORDING TO PED (2014/68/EU) BY CERTIFICATION BODY FOR PRESSURE EQUIPMENT OF TÜV NORD SYSTEMS (NOTIFIED BODY REGISTRATION NO. 0045)

6" SORF 150#

PROD.

MECHANICAL PROPERTIES

Sample location : Mid thick-forging	Test Temp :	Sample Size
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Other applicable Specification :: NACE MR 0175 / ISO 15156 - 2015 NACE MR 0103 - 2015 & MATERIAL CONFORMS TO W2/W9/V10

Melting Process : Electrical induction furnace & Argon Oxygen decarburization (ROS) Concast

Heat Treatment : solution Annealed at 1080°C and water Quenched

Dimension
: Conform with the specification (100% inspected)

Surface quality : Satisfactory Roughness value - RZ 12.5 TO 25 μm /RA 3.2 TO 6.3 μm

PNII Test : No objection (100% tested with mobile spectro)

Corrosion Test : Passed IGC test in accordance with ASTM A262 practice E at micro after IGC (250X)

Micro Observation : No carbide precipitation observed on grain boundaries

Radioactivity Test : We hereby certify that all the material is free from radioactive contamination

Mercury Contamination : Free from contamination

Ultrasonic Test : No objection tested as per ASTM A388 - Marking : VIRAL LOGO, SIZE, TYPE, CLASS, B16.5, MAT. SPEC. Heat No

We certify that the above material has been inspected and tested complies with the order/contract and is of Indian origin



AUTHORIZED WORKS INSPECTOR

SAURABH JAIN



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(A02) INSPECTION CERTIFICATE & MILL TEST REPORT - EN 10204-3.1

(A06)CUSTOMER :		Manufacturer's		(A03) MTR NO.	67215674/9											
		Symbol (A04)		(Z02) DATE	18.01.2024											
				MATERIAL SPEC.												
ASTM A182/182M-17/ASME SA182/SA182M-15																
				ASME SECT.II PART A-2015												
		(B02) GRADE		ASTM A182 UNS S31803/F51												
(B01) STAINLESS STEEL FORGED FLANGES		(B04) DELIVERY CONDITION :		DIMENSIONAL SPEC. ASTM E B16.5-2017												
SO 9001:2015 TÜV NORD REG.No.04109011117M EXPIRY DATE: 21.01.2019 & APPROVED ACCORDING TO AD 2(M) MERRILLATT VM & CERTIFIED ACCORDING TO PED (2014M/ED) BY CERTIFICATION BODY FOR PRESSURE EQUIPMENT OF TÜV NORD SYSTEMS (NOTIFIED BODY REGISTRATION NO.-0045)																
(A07)ORDER NO: 562196		(A08) Sales Order No. 43215674/00452		(A09) Article No:												
(B09-B11) ITEM DESCRIPTION)			QUANTITY (PIECES)		HEAT NUMBER											
S" SORF 150#			100		716392											
(C71-C92) CHEMICAL ANALYSIS																
ELEMENT	%C	%Mn	%SI	%P	%S	%Cr	%Ni	%Mo	%N							
MIN	0.030	2.00	1.00	0.030	0.020	23.00	6.50	3.50	0.20							
MAX	0.022	1.14	0.49	0.025	0.009	21.32	5.27	3.14	0.15							
HEAT																
PROD.																
MECHANICAL PROPERTIES							Test specification ASTM-A370									
Sample location : Mid thick-forging		Test Temp : RT		Sample Shape Round				ASTM E10								
Test Values	(C12) Tensile Strength	(C11) Yield Strength		(C13) Elongation% Lo=4D	Reduction of Area	(C33) Hardness HBW	(C42) Single values		(C43) AVG							
		Rp=0.2%	Rp=1%													
Req.	620 MIN	450 MIN	MPa	MPa	-	290										
T	697.25	576.34	MPa	MPa	38.42	-	243	191	193	195	193					
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